

Macroeconomics

Technological Progress and Growth

(Blanchard, Macroeconomics 8th edition Ch. 12-13)

Technological Progress

- ▶ Technological progress can lead to:
 - ▶ larger quantities of output for given quantities of capital and labor
 - ▶ better products
 - ▶ new products
 - ▶ a large variety of products
- ▶ Think about the services that goods can provide: more services the better.
- ▶ e.g., better car provides more safety; smartphone provides more communication services

Production Function

- ▶ The state of technology (A) is a variable that tells us how much output can be produced from given amounts of capital and labor at any time:

$$Y = F(K, N, A)$$

- ▶ Technology may reduce the number of workers needed, or equivalently, increase output given the number of workers

$$Y = F(K, AN)$$

- ▶ AN is the amount of *effective labor*.
- ▶ Such technical progress is known as labor-augmenting (a.k.a. Harrod-neutral PF).
- ▶ If instead $Y = AF(K, N)$, this is referred to as a Hicks-neutral PF.

Constant Returns to Scale

- ▶ Recall, with constant returns to scale and a given state of technology (A), if the amounts of capital and labor changes by x times, output changes by x times:

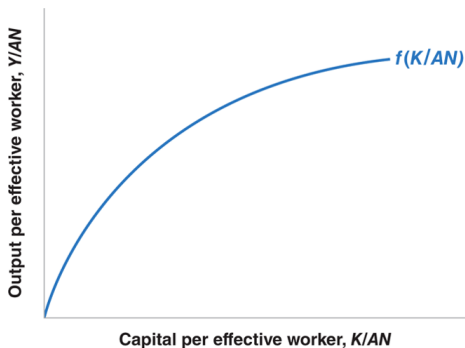
$$F(xK, xAN) = xY$$

- ▶ If $x = 1/AN$, output per effective worker is a function of capital per effective worker:

$$\frac{Y}{AN} = f\left(\frac{K}{AN}\right)$$

- ▶ As before, assume decreasing returns.

Production Function



- Because of decreasing returns to capital, increases in capital per effective worker lead to smaller and smaller increases in output per effective worker.

Solow Model

- ▶ Now, examine dynamics of output and capital per effective worker.
- ▶ As before, if we assume $I = S = sY$, then

$$\frac{I}{AN} = s \frac{Y}{AN} = sf \left(\frac{K}{AN} \right)$$

- ▶ What is the level of investment per effective worker needed to maintain a given level of capital per effective worker K/AN ?

$$I = (\delta + g_A + g_N) K$$
$$\frac{I}{AN} = (\delta + g_A + g_N) \frac{K}{AN}$$

- ▶ where δ is the capital depreciation rate, g_A is the rate of technological progress, and g_N is the rate of population growth.

Solow Model (cont.)

- Using math, first define $k \equiv \frac{K}{AN}$ (note: $\Delta x_{t+1} = x_{t+1} - x_t$ here)

$$\frac{\Delta k_{t+1}}{k_t} = \frac{\Delta K_{t+1}}{K_t} - (g_A + g_N)$$

$$K_{t+1} - K_t = I_t - \delta K_t$$

$$\frac{\Delta k_{t+1}}{k_t} = \frac{I_t}{K_t} - (\delta + g_A + g_N)$$

- In steady state, we must have

$$\frac{I_t}{K_t} = (\delta + g_A + g_N) \Rightarrow I_t = (\delta + g_A + g_N)K_t$$

Dynamics of Capital and Output per Effective Worker

Steady State

- ▶ The steady state of the economy is such that capital per effective worker and output per worker are constant.
- ▶ They are equal to $(K/AN)^*$ and $(Y/AN)^*$, respectively.
- ▶ When the economy is in steady state, output grows at the rate of technological progress and population growth $g_A + g_N$.
 - ▶ If the economy tried to sustain output growth in excess of $g_A + g_N$, capital would have to increase faster than output (decreasing returns), and a larger and larger proportion of output would need to be devoted to capital accumulation.
 - ▶ Independent of saving rate!
- ▶ A measure of the standard of living is output *per worker*, which grows at rate g_A (rate of technological progress).

The Characteristics of Balanced Growth

- ▶ In steady state, output and the two inputs (capital and effective labor), grow “in balance” at the same rate: **balanced growth path**

		Growth rate:
1	Capital per effective worker, K/AN	
2	Output per effective worker, Y/AN	
3	Capital per worker, K/N	
4	Output per worker, Y/N	
5	Labor, N	
6	Capital, K	
7	Output, Y	

The Characteristics of Balanced Growth (con.)

- ▶ On the balanced growth path (steady state or long run):
 - ▶ Capital per effective worker and output per effective worker are constant.
 - ▶ Capital per worker and output per worker are growing at the rate of technological progress.
 - ▶ Capital and output are growing at a rate equal to the sum of population growth and the rate of technological progress.

Effects of an Increase in the Saving Rate

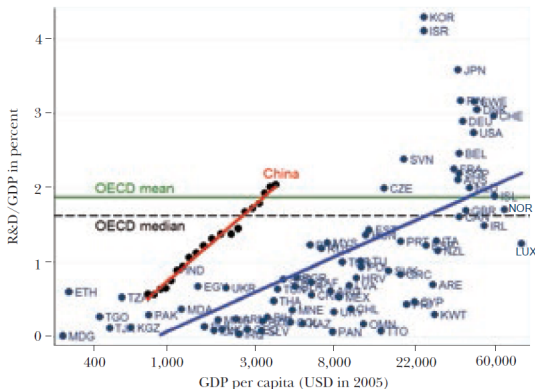
The Determinants of Technological Progress

- ▶ Mankind has made huge discoveries, especially in science:
 - ▶ e.g., invention of microchip, discovery of the structure of DNA
- ▶ But science and economic forces are linked!
- ▶ Most technological progress is the outcome of firms' research and development (R&D) activities.
- ▶ Industrial R&D expenditure accounts for around 2-3% of GDP in countries like US, France, Japan, and the UK.
- ▶ About 75% of the 1 million scientists and researchers in the US are employed by firms (e.g., pharmaceutical companies, I.T. companies).
- ▶ US firms' R&D spending is 20% of the spending on gross investment, 60% on net investment.

R&D Intensity across Countries

Figure 2

Research and Development Intensity Comparison



Source: World Bank, OECD database (<https://data.oecd.org/rd/gross-domestic-spending-on-r-d.htm>).

Note: Data for China are from 1995 to 2014, and data for all other countries are for 2014 or the latest year available.

R&D in Denmark

Vestas to build a large Research- and Development Centre in Aarhus, Denmark

The world's leading manufacturer of wind power systems has decided to build two new technology centres – one in Aarhus, one in Asia – as part of the company's R&D activities.

To cement its leading global market position, the world's largest wind turbine manufacturer Vestas now concentrates its Danish R&D activities in Aarhus. The new 12,500 square meter high-end technology centre will be built on a 25,000 square meter site in Skejby. In time the centre will create up to 500 high technology jobs. Vestas plans for a similar technology centre in Asia – the exact location has yet to be decided.

- Statistics: <https://www.dst.dk/en/Statistik/emner/uddannelse-og-forskning/forskning-udvikling-og-innovation/forskning-og-udvikling>

R&D in Denmark (con.)

[https://www.dr.dk/nyheder/penge/
populaer-fedme-og-diabetesmedicin-giver-novo-nordisk-tocif](https://www.dr.dk/nyheder/penge/populaer-fedme-og-diabetesmedicin-giver-novo-nordisk-tocifret)

PENGE

Populær fedme- og diabetesmedicin giver Novo Nordisk tocifret milliardoverskud

Novo Nordisk hæver sit overskud med hele 46 procent i årets andet kvartal sammenlignet med sidste år.



Outcomes of R&D

- ▶ By increasing spending on R&D, a firm increases the probability that it will discover ideas and develop a new product.
- ▶ Unlike a machine, an idea can potentially be used by many firms at the same time.
- ▶ The level of R&D spending depends not only on the *fertility of research* (how spending on R&D translates into new ideas and new products),
- ▶ But also on the *appropriability* of research results (the extent to which firms can benefit from the results of their own R&D).

Fertility of Research

- ▶ Many factors may determine the fertility of the research process.
- ▶ Basic versus applied research:
 - ▶ Basic: search for general principles and results (e.g., physicist)
 - ▶ Applied: application of results to specific uses, development of new products (e.g., engineering)
- ▶ France: abstract thinking; US: culture of entrepreneurship
- ▶ e.g., progress in encryption comes from theory of prime numbers
 - ▶ <https://www.abc.net.au/news/science/2018-01-20/how-prime-numbers-rsa-encryption-works/9338876>
 - ▶ <https://www.independent.co.uk/news/take-one-blackboard-one-piece-of-chalk-1257787-twos-and-wor.html>
- ▶ Will research become less fertile?

Foreign Direct Investment (FDI)

- ▶ The availability of capital is clearly one key economic benefit brought forward by FDI to the host economy.
- ▶ FDI contributes to economic growth through the diffusion and assimilation of technologies and technological know-how.
- ▶ It has been noted that MNCs may spread technological change through the acquisition of foreign firms and not through “Greenfield” projects only.
- ▶ FDI may also affect technological change through productivity spillovers.
- ▶ Huawei: <https://www.abc.net.au/news/2019-01-29/huawei-technologies-indicted-bank-fraud-stealing-trade-secrets/10757594>

Appropriability of Research Results

- ▶ If firms cannot appropriate the profits from the development of new products, they will not engage in R&D.
- ▶ Company may have trade secret (e.g., Coca-Cola). But usually, companies need legal protection.
 - ▶ **Patents** give a firm that has discovered a new product (i.e., technique or device) the right to exclude anyone else from the production or use of that new product for some time.
- ▶ Trade-off: Protection is needed to provide firms with the incentives to spend on R&D.
 - ▶ e.g., biogenetic research
- ▶ But it would be best for society if knowledge is made available without restriction.
 - ▶ Open-source software? Should students be allowed to download textbooks?

Management Practices: Another Dimension

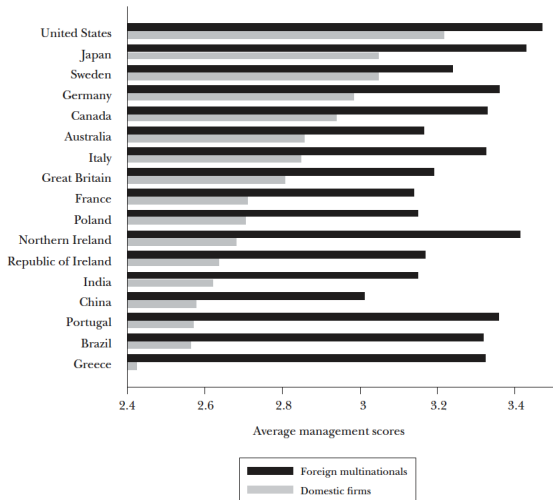
- ▶ Some researchers believe that management practices might be stronger than many of the other factors that determine a firm's performance, including technological innovations.
- ▶ Two economists (Nick Bloom, John Van Reenen) studied management practices and performance of more than 4,000 medium-sized manufacturing operations in Europe, the US and Asia.
- ▶ They found that firms used the same technology but applied good management practices perform significantly better than those that did not.

Management Practices (con.)

- ▶ In another study, they conducted a field experiment:
- ▶ Free consulting to Indian textile plants (out of 20), good management practices raised productivity by 18% through improved quality and reduced inventory.
- ▶ <https://voxdev.org/topic/firms-trade/do-management-interventions-last>
- ▶ <https://voxdev.org/topic/firms/do-management-interventions-last-evidence-india>
- ▶ See also <https://cepr.org/voxeu/columns/adding-piece-productivity-puzzle-management-practices>

Management Practices (con.)

Multinationals Are Well Managed in All Countries



Source: Bloom, Genakos, Sadun, and Van Reenen (2009).

Innovation and Imitation

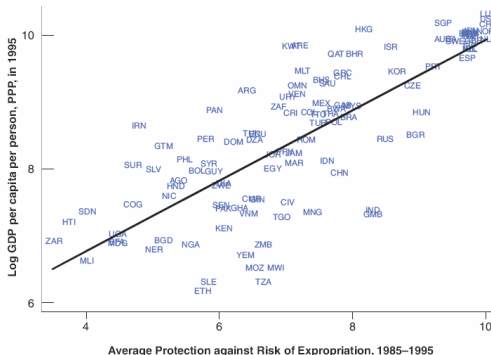
- ▶ To sustain growth, advanced countries at the technology frontier must innovate.
- ▶ Poorer countries further from the technology frontier can instead grow largely by
 - ▶ Imitating rather than innovating
 - ▶ Importing and adapting existing technologies instead of developing new ones
- ▶ Firms must pay fee or royalties to use foreign technology
<https://www.automobilemag.com/news/hyundai-engines/>
- ▶ e.g., China: importation and adaption has been important, and may explain the poor enforcement of patent rights.
- ▶ China vs. Africa? Development economics.

Institutions, Technological Progress, and Growth

- ▶ Proximate causes: investment, capital accumulation, human capital, technology
- ▶ Fundamental causes: geography (e.g., soil, topography), culture (e.g., values, religion), institutions (e.g., regulations, laws)
- ▶ Jared Diamond: “Guns, Germs, and Steel: The Fates of Human Societies”
 - ▶ Why is it that Europeans conquered the Americas, most of Africa, Australia, etc?
 - ▶ Why didn't the Incas of Peru, or the Aztecs of Mexico, or the Ashanti of Ghana conquer Europe instead?

Expropriation and Income

- Expropriation: profits appropriated by state, extracted in bribes by corrupt bureaucrats, stolen by others.



- There is a strong positive relation between the degree of protection from expropriation (i.e., of property rights) and the level of GDP per person.

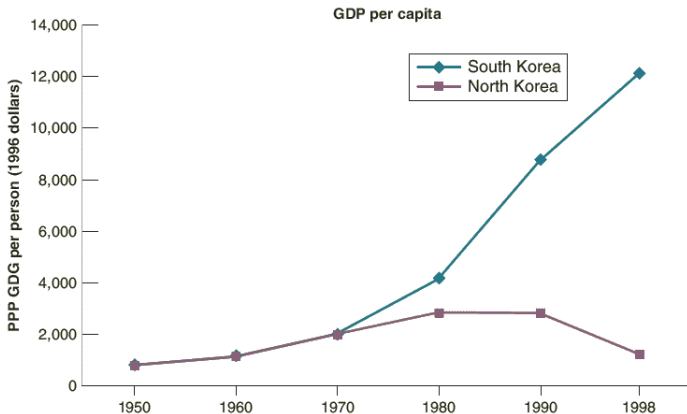
Protection of Property Rights

- ▶ May entail a good political system, in which government cannot seize or expropriate the property of citizens
- ▶ Good judicial system, where disagreements can be resolved efficiently and fairly.
- ▶ Laws, such as against insider trading in the stock market, patent laws, antitrust laws against monopolies.
- ▶ World Bank World Governance Indicators:
<http://info.worldbank.org/governance/wgi/>

North Korea and South Korea

- ▶ After WWII, Korea formally acquired independence but was divided into two zones of occupation:
 - ▶ Soviet armed forces in north
 - ▶ US armed forces in south
- ▶ Korean War lasted from 1950 to 1953, with armistice in 1953.
- ▶ Ethnic and linguistic homogeneity (culture, religion), economically similar → natural experiment!
- ▶ South Korea relied on capitalist organization of economy, with strong state intervention but also private ownership and legal protection of private producers.
- ▶ North Korea relied on central planning, nationalization of industries, no private property rights → decline in agriculture and industry.

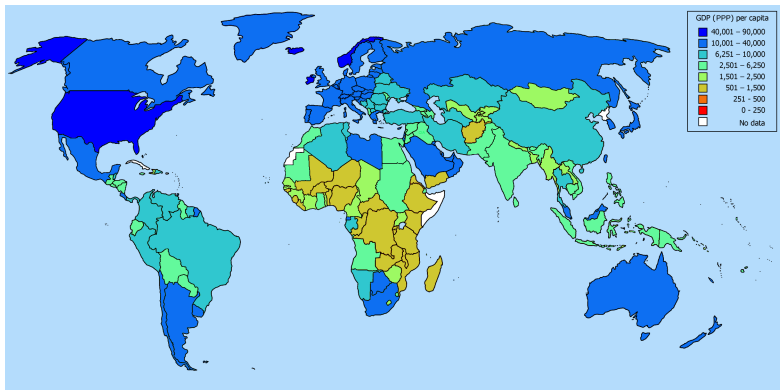
North Korea and South Korea



- 50 years later, GDP per person was 10x higher in South Korea.

The Role of Institutions

- ▶ Remarkable correlation between latitude and economic prosperity.



- ▶ Nations closer to equator have lower levels of income per person.
- ▶ One explanation: tropical climates have direct negative impact on productivity, e.g., agriculture more difficult, disease more prevalent.

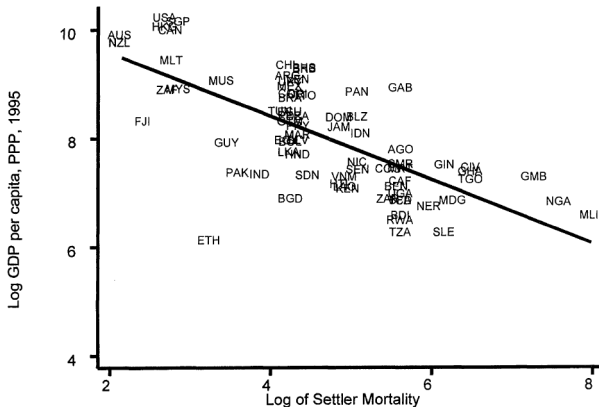
Explanation of Acemoglu et al. (2001)

- ▶ In 17th, 18th, 19th C., tropical climates presented European settlers with an increased risk of disease, e.g., malaria, yellow fever.
- ▶ When Europeans were colonizing, they avoided settling in tropical areas (e.g., Africa, Central America).
 - ▶ Set up “extractive” institutions for natural resources, e.g., authoritarian governments
- ▶ Preferred areas with more moderate climates and better health conditions (e.g., US, Canada, New Zealand).
 - ▶ Set up European-like institutions that protected individual property rights and limited the power of government

Explanation of Acemoglu et al. (2001) (con.)

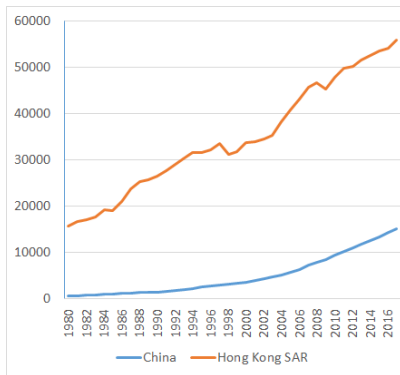
- ▶ Early institutions are strongly correlated with the modern institutions in former colonies.
 - ▶ Extractive institutions remained and taken over by new ruling elites.
- ▶ Quality of institutions is key to economic performance.
 - ▶ Where property rights are well protected, people have more incentive to make the investments that lead to economic growth.

Settler Mortality and Income

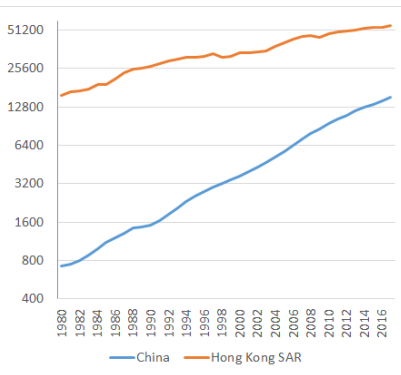


- Mortality rates of soldiers, bishops, and sailors stationed in colonies between 17th and 19th C., largely based on historian Philip Curtin.

Hong Kong and China's GDP per capita



(a)



(b)

<https://www.youtube.com/watch?v=zY2iPePjXAU>

What Is Behind Chinese Growth?

- ▶ From 1949 to late 1970's, China relied on central planning.
- ▶ Output decreased by 20% from 1959-1962, and 10% from 1966-1968.
- ▶ Agricultural reform in 1978 allowed farmers, after satisfying quota from state, to sell production in rural markets.
- ▶ Over time, farmers obtained increasing rights to their land, and today, state farms produce less than 1% of agricultural output.
- ▶ State firms were given increasing autonomy over production decisions, and market mechanisms and prices were introduced.
- ▶ Private entrepreneurship was encouraged, tax advantages and special agreements were used to attract foreign investors (e.g., from Hong Kong).

What Is Behind Chinese Growth?

- ▶ Average growth of output per worker in China has increased from 2.5% between 1977, to more than 9% since then.
- ▶ Unlike Central and Eastern Europe, China's state sector has declined slowly and that decline has been more than compensated by strong private sector growth.
- ▶ Also, the Chinese political system did not change, and the government was able to control the pace of transition to a market economy.
- ▶ As property rights are still not well established and the banking system is still inefficient, the limits of Chinese growth are clear.

Constructing a Measure of Technological Progress

Solow Residual

- ▶ In 1957, Robert Solow devised this way to construct an estimate of technological progress.
- ▶ The Solow residual, i.e., rate of growth of total factor productivity (TFP), measures the effects of technological progress.
- ▶ Different from rate of growth of labor productivity, $g_Y - g_N$.

$$\text{residual} = \alpha g_A \Rightarrow g_A = \text{residual} / \alpha$$

- ▶ e.g., $\alpha = 0.6$, $g_N = 1.7\%$, $g_K = 9.2\%$, $g_Y = 7.2\%$.

Growth in Rich Countries since 1985

- ▶ According to our theories, fast growth may come from 2 sources:

- 1.

- 2.

- ▶ How do we tell them apart?

Growth in Rich Countries since 1985 (con.)

- Average Annual Rates of Growth of Output per Worker and Technological Progress in Four Rich Countries since 1985

	Rate of Growth of Output per Worker (%) 1985–2014	Rate of Technological Progress (%) 1985–2013
France	1.3	1.4
Japan	1.6	1.7
United Kingdom	1.9	1.4
United States	1.7	1.4
Average	1.6	1.5
Source: Calculations from the OECD Productivity Statistics.		

Growth in Rich Countries since 1985 (con.)

- ▶ Over the period 1985-2014, output per worker has grown at rather similar rates across the five countries.
 - ▶ Much of the convergence happened between 1950 and 1985, and appears to have slowed down or stopped since then.
- ▶ Growth since 1985 has mostly come from technological progress, not from unusually high capital accumulation.
 - ▶ Column 1 roughly same as column 2 $\Rightarrow$ appears to be on balanced growth path.
 - ▶ Capital accumulation allowed countries to achieved balanced growth, but growth did not come from an “unusual” increase in capital accumulation.

Growth in China

- ▶ Growth of Output per Worker and Technological Progress:

Period	Rate of Growth of Output (%)	Rate of Growth of Output per Worker (%)	Rate of Technological Progress (%)
1978–1995	10.1	7.4	7.9
1996–2011	9.8	8.8	5.9

Source: Penn World Table version 8.1.

- ▶ Over the period 1978-1995, China was on a balanced growth path as the rate of technological progress was close to the rate of growth of output per worker.
- ▶ Since 1996, although growth of output per worker has remained high, the contribution of technological progress has decreased and we have unusually high capital accumulation.
- ▶ Productivity growth due to labor transferring from the countryside to cities, and imported technology from more technologically advanced countries (e.g., joint ventures).

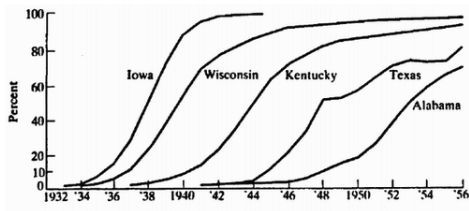
The Challenges of Growth

- ▶ We have argued that in the long run, technological progress is the key to increases in the standard of living.
- ▶ Yet technological progress is often blamed for higher unemployment, higher income inequality, and more damage to the environment.
- ▶ Some economists believe that the era of major inventions is largely over, and we should expect technological progress to slow down considerably.
- ▶ Others believe instead that we are on the verge of another technological revolution and the future is bright.
- ▶ (What about Artificial Intelligence (AI)?)

The Future of Technological Progress

- ▶ One possible reason that we see such rapid advances in technology that appear to have little impact on productivity growth is due to measurement error.
- ▶ Measuring technological progress/productivity is difficult. For example, how much better is a 2018 computer relative to a 2010 computer? Are we understating the rate of productivity growth?
- ▶ If productivity growth has declined it might be explained by the magnitude of innovation.
- ▶ Some argue that the current major innovations are less important than the major innovations of the past.
- ▶ Major innovations are innovations that have applications in many fields and many products: *general purpose technologies*, e.g., electricity, internal combustion engine.

Technological Diffusion

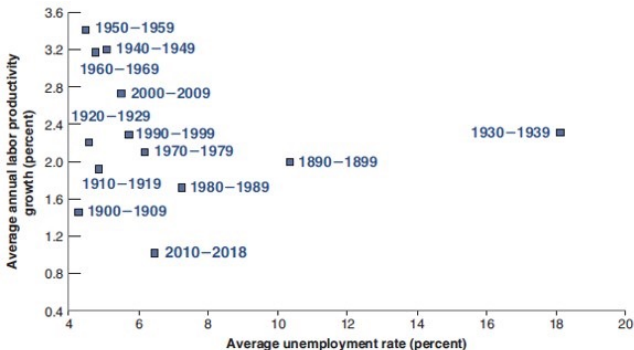


- ▶ Percentage of total corn acreage planted with hybrid seed
- ▶ Griliches showed that the speed of adoption was a function of the profitability of introducing hybrid corn.

Robots and Unemployment

- ▶ Since the Industrial Revolution workers have worried about being replaced by technology.
- ▶ The theme of *technological unemployment* typically resurfaces whenever unemployment is high.
- ▶ However, the argument that technological progress must necessarily lead to higher unemployment is obviously false.
- ▶ The large improvements in the standard of living that advanced countries enjoy have come with large increases in employment and no increase in the unemployment rate.
- ▶ These increases in the standard of living coincided with increasing productivity. More sophisticated versions of the technological unemployment hypothesis, however, must be taken more seriously.

Productivity Growth and Unemployment, 1890-2018



- Source: Data prior to 1960: Historical Statistics of the United States. Data after 1960: Bureau of Labor Statistics.

Technology and Unemployment

- ▶ However, this past relationship may not hold in the future.
- ▶ By their very nature, they can replace the least skilled workers.
- ▶ A recent study (by Acemoglu and Restrepo) concludes that, over the next two decades, 47% of US workers are at risk of being replaced by robots.
- ▶ It is possible that robots may create other jobs to replace the ones eliminated.
- ▶ To date, robots have not led to higher unemployment.
- ▶ <https://www.youtube.com/watch?v=DPyzVtMs-Vo>
- ▶ <https://www.youtube.com/watch?v=Em06dBcqQ2w>

More on Robots

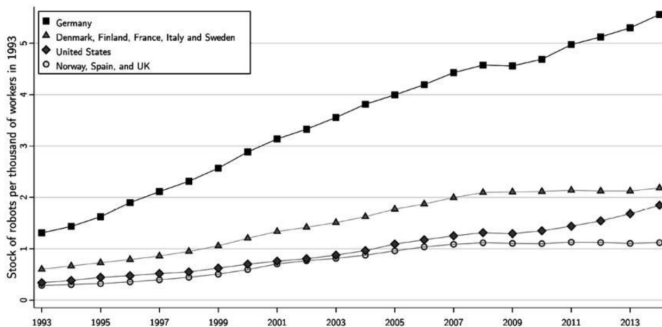
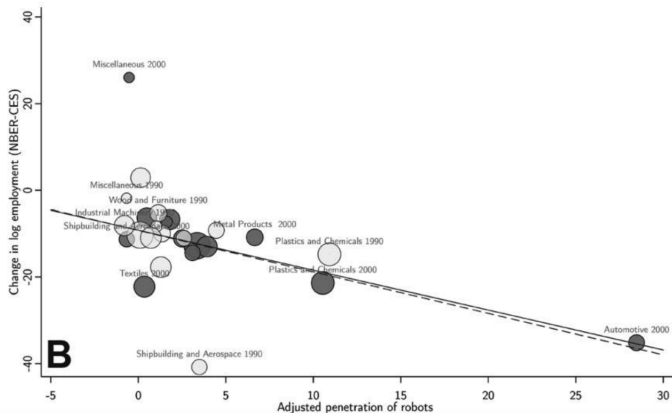
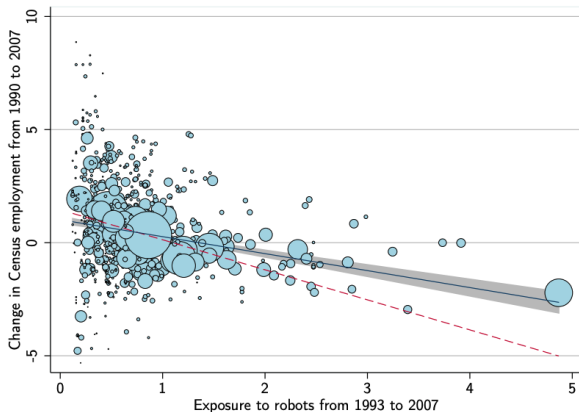


FIG. 1.—Industrial robots per thousand workers in the United States and Europe.

Robots and Employment: Industries



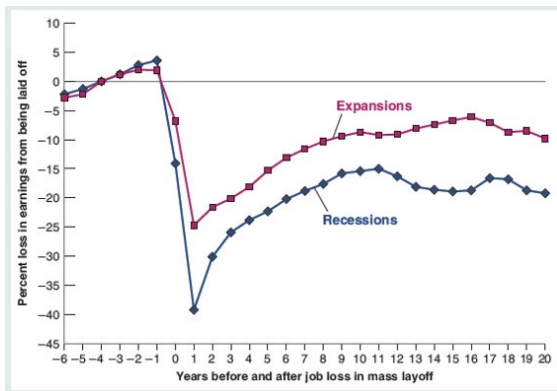
Robots and Employment: Commuting Zones



Growth and Churn

- ▶ Joseph Schumpeter emphasized the process of growth was fundamentally a process of *creative destruction*: New goods are developed, making old ones obsolete.
- ▶ New techniques of production are introduced, requiring new skills and making some old skills less useful. As this happens, old jobs are eliminated, new jobs are created.
 - ▶ e.g., ATM has replaced a teller, but engineers are needed to make the ATM
 - ▶ Today, there are much fewer farmers, more car drivers, programmers, etc.
- ▶ Even for those with the right skills, the firm where they work may be replaced by a more efficient firm, or the product their firm was selling may be replaced by another product.

Job Destruction, Churn and Earnings Losses

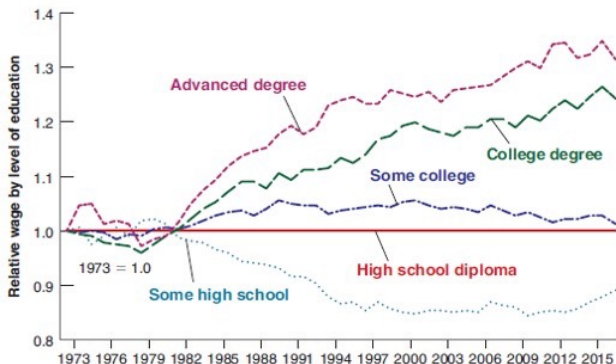


Source: Steven J. Davis and Till M. von Wachter, "Recessions and the Cost of Job Loss," National Bureau of Economics Working Paper No. 17638, 2011. Used with Permission.

Job Destruction, Churn and Earnings Losses (con.)

- ▶ Using records from the Social Security Administration between 1974 and 2008, the study looks at what happens to workers who lose their job as a result of a mass layoff.
- ▶ Recall that in Topic 1 we looked at life satisfaction for unemployment.
- ▶ Technological change is the main source of growth in the long run.
- ▶ It clearly enables a higher standard of living for the average person.
- ▶ But workers who experience mass layoffs are still clear losers.
- ▶ Mass layoff, if due to technological change, renders specific skills much less valuable.

Growth and Inequality



- ▶ Since the early 1980s, the relative wages of workers with a low education level have fallen; the relative wages of workers with a high education level have risen.

Growth and Inequality (con.)

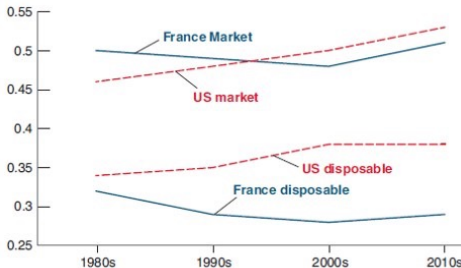
- ▶ In growing sectors or occupations, or for those with the right skills, technological progress leads to opportunities and higher wages.
- ▶ But for the jobs in which labor demand is low, they may be unemployed or suffer low wages.
- ▶ The primary cause of the increase in wage inequality is the shift in demand for skill sets. Two possible reasons for this shift are:
 1. **International trade and offshoring:** Firms that employ higher proportions of low-skill workers, are increasingly driven out of markets by imports from similar firms in low-wage countries.
 2. **Skill-biased technological change (SBTC):** New machines and new methods of production require more and more high-skill, high-education workers.

Income and Wealth Inequality

- ▶ <https://www.youtube.com/watch?v=41y4c10i5Uo>
- ▶ Is it possible for countries to sustain growth without having ever-increasing inequality?
- ▶ There are two reasons for optimism:
 1. Not all advanced economies are like the US. Wage inequality and the share of the top 1% have been rising in most countries, but typically much less than in the US.
 2. We must distinguish between *market income* inequality, which is income inequality before taxes and transfers, and *disposable income* inequality, which is income after taxes and transfers.

Gini Coefficient

Evolution of Gini Coefficients



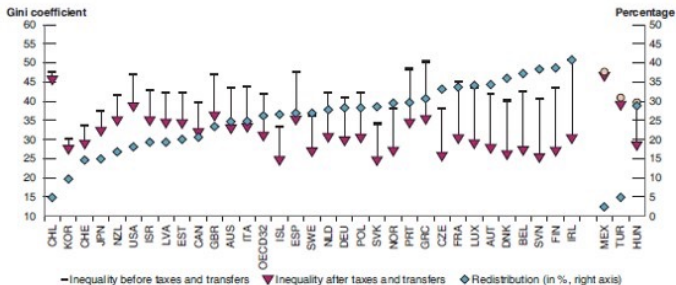
Source: Income Inequality by Max Roser and Esteban Ortiz-Ospina, First published in December 2013; updated October, 2016.

(<https://ourworldindata.org/income-inequality>).

Reducing Inequality

- ▶ There are strong moral reasons to want to limit inequality, to achieve *inclusive growth* (in China, “common prosperity”.)
- ▶ There are also good political reasons for governments to do so: Perceived inequality has played a role in the rise of populism.
- ▶ To reduce inequality governments can try to reduce market income inequality, through better education, through minimum wages, and through rules about governance within firms.
- ▶ Or, given market income inequality, they can reduce disposable income inequality through progressive taxation, including the use of a negative income tax to increase revenue at the low end of the wage distribution
- ▶ Also through transfers, e.g., food stamps, unemployment benefits.

Redistribution across OECD Countries



Source: Market income, Disposable income, and redistribution, across OECD countries. Used with Permission.

Income redistribution across OECD countries: main findings and policy implications.

Climate Change and Global Warming

- ▶ Markets work poorly when there are externalities.
- ▶ A major externality of growth is the emission of greenhouse gases.
- ▶ Since the Industrial Revolution, the use of fossil fuels (primarily coal) has led to a large increase in CO₂ emissions.
- ▶ At the same time, there has been a steady increase in the average global temperature.
- ▶ Strong evidence for each, therefore, at the very least, there is positive correlation.
- ▶ China is now the main source of CO₂ emissions (coal), but they argue that Europe and the US were able to grow and emit earlier in time.